



Investigation of Nonlinear Cylindrical Electrostatic Excitations in Dense Quantum Astrophysical Plasmas

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Abstract

It is imperative for many problems of physical interest to incorporate the geometrical curvature. Examples include plasma physics, oceanography, nonlinear optics, and laser-driven systems. Therefore, we consider planar wave propagation in a cylindrical geometry in light of the aforementioned applications, and the propagation is considered solely in the radial direction. Using the small amplitude perturbation approximation, the cylindrical Korteweg-de Vries (CKdV) equation is obtained using multiple-scale analysis to study nonlinear ion-acoustic waves in a dense plasma with electron trapping by incorporating the effects of the quantizing magnetic field and the smearing effects of the Fermi distribution function. The Bäcklund transformation is employed to obtain single and multiple soliton solutions of the CKdV equation, which are found to be very different from the planar KdV equation. A general mathematical framework is also presented to find the N-soliton solutions. The effects of the quantizing magnetic field and finite electron temperature on the structure of the cylindrical ion-acoustic solitons are also explored using the parameters representative of white dwarf stars. This research endeavor is expected to trigger interest in the plasma community to pursue this fascinating and abstruse research direction.

Keywords Bäcklund transformation · Cylindrical Korteweg-de Vries equation · Astrophysical Plasmas · N-soliton solutions

1 Introduction

For a system whose temperature is lower than the Fermi temperature, it becomes imperative to ascertain the behav-

ior of its constituent particles using the laws of quantum mechanics. In dense plasmas, the tenuous electrons, which are three orders of magnitude less massive than ions, are studied quantum mechanically, whereas the gigantic ions are treated like classical particles [1, 2]. Such systems have garnered a lot of attention due to their critical and exciting applications in many disciplines of physics like solid-state physics [3–5], nanowires [6, 7], nanostructures formed by quantum plasmons [8–10], interaction of lasers with matter [11], experiments in inertial confinement in the context of fusion [12–14], to comprehend the structure of stars in astrophysics [15–17], and dense structures, namely, white dwarfs and neutron stars [18–20]. Different wave modes emanate in the plasma because of the electric and magnetic fields and their collective interactions [1, 21]. Note that the physical model that is most frequently employed to solve a system of charged particles that are dictated by different statistical laws due to the difference in their inertia is the quantum hydrodynamical (QHD) model [22–24]. A somewhat helpful description of this model is presented by the Wigner-Poisson system [25]. Manfredi and Haas [26] used this model to explain linear and nonlinear waves in a plasma [27] whereas Tsintsadze et al. [28] employed the Fermi-Dirac statistics and

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Pauli equations in the non-relativistic regime to investigate the wave propagation in quantum plasmas.

Gurevich made the early advances in the problem of electron trapping in classical plasmas [29]. The author solved the trapping problem by assuming a weak field. Decades later, Shah et al. [30] gainfully used the work carried out by Gurevich to examine quantum ion-acoustic waves (IAWs). This work led to significant advances in the field of electron trapping in quantum plasmas by studying the arbitrary amplitude nonlinear electrostatic waves on the ion time scale for a relativistically degenerate gas [31] and a quantizing magnetic field (QMF) [32]. The initial study of electron trapping in a quantizing magnetic field for a non-relativistically degenerate gas was later extended by Iqbal et al. for a relativistically degenerate gas [33]. Electrostatic waves were studied in dense plasmas in linear and nonlinear regimes with quantum mechanically trapped particles for spatially non-uniform plasmas [34]. Later, the influence of Landau quantization (LQ) on the structure of nonlinear drift waves was examined [35], and it was also shown that the presence of finite electron temperature (FET) led to the formation of both humps and cavities [36]. A comprehensive literature review of electron trapping in quantum plasmas to study nonlinear electrostatic waves was written by Masood et al. [37], which presented the details of the works pursued in this direction in the last fifteen years. The problem of electron trapping has also been investigated for quantum shock waves in spatially inhomogeneous plasmas and kinetic Alfveń waves and in classical plasmas for distribution functions that exhibit a departure from Gaussian profile [38–40]. The influence of the QMF and temperature has been explored for semiconductor materials at the nanoscale and in the band gap theory of semiconductors by Erkaev et al. [41–43]. Very recently, Sabeela et al. [44] analyzed the unconventional solutions of the KdV equation and their interaction using the Wronskian approach in quantum plasma. The authors reported solutions with remarkable wave enhancement properties and spelled out possible applications of such solutions in the study of nonlinear wave propagation. Another recent article studied a modified form of the Zakharov-Kuznetsov equation in a spatially non-uniform and collisional plasma in the presence of QFM and partial degeneracy and explored the influence of these parameters on the formation of these structures [45].

In 1880, A.V. Bäcklund pioneered the Bäcklund transformation (BT), a method using first-order partial differential equations to link two distinct functions, allowing them to act as transformations of one another. By starting with a basic solution, the BT facilitates the creation of new solutions for nonlinear PDEs, transforming one equation to another while maintaining solution consistency [46], thus offering a powerful tool for tackling complex nonlinear problems. For a comprehensive overview and detailed discussion on BT, readers should consult our earlier work [46], which in

the framework of the cylindrical KdV equation and with the assistance of the BT method, the cylindrical multi-soliton solutions have been investigated in an unmagnetized collisionless non-Maxwellian plasma comprising inertial ions and inertialess two-temperature electrons following kappa distribution.

Since the nonplanar (cylindrical and spherical) third-order KdV-type equations [47, 48] and the fifth-order KdV-type (Kawahara-type) [49–51] equations are the most accurate for studying realistic phenomena, thus, many researchers have focused on their investigation on this family to study the distinctive properties of different physical phenomena (nonplanar solitary waves, nonplanar cnoidal waves, nonplanar shock waves, etc.) in several plasma systems. Since this family is not integrable, i.e., it does not support an exact solution in its standard form. Thus, many researchers have used different methods to analyze and solve this issue and obtain approximate solutions that simulate natural phenomena well [52–56]. Decades earlier, Nakamura et al. [57] employed the bilinear BT method to discover solutions (both single and two-soliton) of the CKdV equation. In this study, we demonstrate the existence of single-, two-, and three-soliton solutions of the CKdV equation using a distinct methodology that differs significantly from the method employed by Nakamura's team. In addition, we have created a comprehensive mathematical framework to derive solutions involving N -solitons. Unlike the KdV equation in the Cartesian geometry, the solutions we obtain for the CKdV equation distinctively emerge as Airy functions, thereby introducing an intricate dimension to their characterization.

The manuscript proceeds as follows: Sect. 2 elucidates the physical model employed to explore the structure of nonlinear IAWs within a cylindrical framework. Section 3 is dedicated to deriving the CKdV equation using the reductive perturbation technique (RPT). Sections 4 and 5 present the rigorous mathematical details of the BT that are exploited to obtain solutions to the CKdV equation. In Sect. 6, we apply these theoretical principles to investigate electron trapping effects in a degenerate plasma influenced by QFM and FET, using the plasma parameters characteristic of white dwarf stars. The article concludes in Sect. 7 with a concise summary of the research findings.

2 Depiction of the Fluid Model

In our investigation, we utilize the multi-fluid theory to explore the characteristics of a degenerate plasma subjected to a QMF. This plasma is composed of electrons governed by the principles of quantum mechanics. In contrast, ions, due to their significantly larger mass compared to electrons, can be effectively described using Newtonian equations. The wave propagation is assumed to be in the radial direction. In this

case, the equilibrium condition reads $n_{i0} = n_{e0}$, where n_{i0} and n_{e0} indicate the ions and electrons unperturbed number densities, respectively. To examine the behavior of the IAWs within this framework, given below are the set of dimensionless fluid equations

$$\frac{\partial n_i}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r n_i v_i) = 0, \quad (2.1)$$

$$\frac{\partial v_i}{\partial t} + v_i \frac{\partial v_i}{\partial r} = -\frac{\partial \Phi}{\partial r}, \quad (2.2)$$

and

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \Phi}{\partial r} \right) = n_e - n_i. \quad (2.3)$$

Building upon the established framework for Fermi integrals and employing the methodology explained in detail in Refs. [30, 31], the expression for the total number density of electrons, taking into account Landau quantization and the smearing effects of the Fermi distribution function, is obtained as follows:

$$n_e = N_0 \left[\frac{3}{2} \eta (1 + \Phi)^{1/2} + (1 + \Phi - \eta)^{3/2} - \frac{\eta T^2}{2} (1 + \Phi)^{-3/2} + T^2 (1 + \Phi - \eta)^{-1/2} \right] \quad (2.4)$$

Expanding (2.4) under the condition that $\Phi < 1$ yields the following expression:

$$\frac{n_e}{n_{i0}} = \left[1 + A_1 \Phi + A_2 \Phi^2 \right], \quad (2.5)$$

with $A_1 = \frac{A_2}{A_0}$ and $A_2 = \frac{A_3}{A_0}$, where the coefficients A_0 , A_2 , and A_3 are defined as follows:

$$A_0 = \frac{3}{2} \eta + (1 - \eta)^{3/2} - \frac{\eta T^2}{2} + T^2 (1 - \eta)^{-1/2}, \quad (2.7)$$

$$A_2 = \frac{3}{2} \left[\frac{\eta}{2} (1 + T^2) + (1 - \eta)^{-1/2} - \frac{T^2}{3} (1 - \eta)^{-3/2} \right] \Phi, \quad (2.8)$$

$$A_3 = \frac{3}{8} \left[-\frac{\eta}{2} (1 + 5T^2) + (1 - \eta)^{-1/2} + T^2 (1 - \eta)^{-5/2} \right] \Phi^2. \quad (2.9)$$

Here, the influence of the QMF is evident through the expression $\eta = \hbar \omega_{ce} / \epsilon_{Fe}$, where $\epsilon_{Fe} = (\hbar^2 / 2m_e) (3\pi^2 N_0)^{2/3}$ represents the electron Fermi energy. $N_0 = p_{Fe}^3 / 3\pi^2 \hbar^3$ denotes the equilibrium number density for a fully degenerate plasma, with m_e representing the mass

of electrons. The Fermi ion sound velocity is given as $c_{s,F} = (\epsilon_{Fe} / m_i)^{1/2}$. The space and time scales are normalized with Thomas-Fermi length which is given as $\lambda_{TF} = (\epsilon_{Fe} / 4\pi e^2 N_0)^{1/2} \equiv c_{s,F} / \omega$ and inverse ion plasma frequency $\omega_{pi}^{-1} = (4\pi n_{i0} Z_i^2 e^2 / m_i)^{-1/2}$, respectively [58]. Also, $v = v_i / c_{s,eff}$, $n_i = n_i / n_{i0}$, $\Phi = e\phi / \epsilon_{Fe}$.

3 Derivation of Evolutionary (CKdV) Equation

This investigation scrutinizes ion acoustic solitary waves (IASWs) characterized by minute yet discernible amplitudes as they propagate radially. Employing the principles of weak turbulence theory, we introduce a methodology to elucidate their behavior. This involves the adoption of stretched coordinates to delineate spatial and temporal variables, exemplified by

$$\xi = \epsilon^{1/2} (r - \lambda t) \text{ and } \tau = \epsilon^{3/2} t, \quad (3.1)$$

where λ denotes the phase velocity of the wave of interest normalized with the quantum ion sound velocity $c_{s,F}$ and ϵ is a diminutive, real parameter ($0 < \epsilon < 1$) that signifies the potency of the nonlinearity. The expansion of the physical variables n_i , v_i , and Φ is given as follows:

$$n_i = 1 + \epsilon n_{i1} + \epsilon^2 n_{i2} + \epsilon^3 n_{i3} + \dots, \quad (3.2a)$$

$$v_i = \epsilon v_{i1} + \epsilon^2 v_{i2} + \epsilon^3 v_{i3} + \dots, \quad (3.2b)$$

$$\Phi = \epsilon \Phi^{(1)} + \epsilon^2 \Phi^{(2)} + \epsilon^3 \Phi^{(3)} + \dots \quad (3.2c)$$

By substituting (3.1) and (3.2a)–(3.2c) into (2.1)–(2.5) and melding the terms in ϵ that appear in the lowest order, we get

$$v_{i1} = \frac{\Phi^{(1)}}{\lambda}, \quad (3.3a)$$

$$n_{i1} = \frac{\Phi^{(1)}}{\lambda^2}, \quad (3.3b)$$

$$n_{i1} = (A_1) \Phi^{(1)}. \quad (3.3c)$$

We simultaneously solve (3.3a)–(3.3c) and arrive at the following linear dispersion relation for our physical model:

$$\lambda = \sqrt{\frac{1}{A_1}}. \quad (3.4)$$

Note that the effects of the physical parameters are encapsulated in A_1 . Collecting the $\epsilon^{3/2}$ -order terms yields the

following set of equations:

$$-\lambda \frac{\partial n_{i2}}{\partial \xi} + \frac{\partial v_{i2}}{\partial \xi} + \frac{v_{i1}}{\lambda \tau} + \frac{\partial n_{i1}}{\partial \tau} = -\frac{\partial (n_{i1} v_{i1})}{\partial \xi}, \quad (3.5a)$$

$$-\lambda \frac{\partial v_{i2}}{\partial \xi} + \frac{\partial v_{i1}^2}{\partial \xi} + \frac{\partial v_{i1}}{\partial \tau} = -\frac{\partial \Phi^{(2)}}{\partial \xi}, \quad (3.5b)$$

$$n_{i2} - A_1 \Phi^{(2)} = -\frac{\partial^2 \Phi^{(1)}}{\partial \xi^2} + A_2 \Phi_1^{(2)}. \quad (3.5c)$$

The simultaneous solution of (3.5a)–(3.5c) results in the CKdV equation

$$\frac{\partial \Phi^{(1)}}{\partial \tau} + C \Phi^{(1)} \frac{\partial \Phi^{(1)}}{\partial \xi} + D \frac{\partial^3 \Phi^{(1)}}{\partial \xi^3} + \frac{\Phi^{(1)}}{2\tau} = 0, \quad (3.6)$$

with

$$C = \frac{3}{2\lambda} - \lambda^3 A_2, \quad (3.7a)$$

$$D = \frac{\lambda^3}{2}. \quad (3.7b)$$

Note that the last term in the above equation owes its presence to the nonplanar geometry. Using the transformations $\Phi^{(1)} = 6C^{-1} D^{1/3} u$, $\xi = D^{1/3} x$, and $\tau = t$ reduce (3.6) to the standard CKdV equation

$$u_t + 6uu_x + u_{xxx} + \frac{u}{2t} = 0. \quad (3.8)$$

For (1+1) dimensional equations, the CKdV equation stands as a quintessential and, therefore, extremely important. Initially examined by Maxon and Vieceilli [57, 59, 60], this equation was found to admit infinite conservation laws [61].

4 Bäcklund Transformation (BT) of the CKdV Equation

Two solutions z and z' of the CKdV equation are expressed by $u = -z_x$, $u' = -z'_x$ or $z = -(2 \log f)_x$, $z' = -(2 \log f')_x$ [57]. It is worth noting that the prime notation here does not denote differentiation. Therefore, the BT for the CKdV

equation can be represented in the following manner [57]:

$$(Z'_+)_x = \frac{1}{2}(-Z'_-)^2 - \frac{x + x_1}{6t}, \quad (4.1a)$$

$$(-Z'_-)_t = (Z'_-)_{xxx} - \frac{3}{2}\{(Z'_-)(Z'_+)_x\}_x + \frac{1}{4}\{(Z'_-)^3\}_x + \left\{\frac{x + x_1}{4t}(Z'_-)\right\}_x, \quad (4.1b)$$

where $Z'_- \equiv (z - z')$ and $Z'_+ \equiv (z + z')$.

At this juncture, the constant x_1 , arbitrarily selected, carries a notable physical significance as the initial position of the soliton [57]. It is easily verifiable that the system of (4.1a) and (4.1b) adheres to the integrability condition, expressed as (i.e., $(Z'_+)_{xt} = (-Z'_-)_{tx}$). Initiating from the vacuum solution $z' = 0$ and $x_1 = 0$ within (4.1a), (4.1b), we derive the subsequent equations

$$\frac{dz}{dx} = \frac{1}{2}z^2 - \frac{x}{6t}, \quad (4.2)$$

$$\frac{dz}{dt} = -\frac{d^3 z}{dx^3} + \frac{3}{2}\left(\frac{dz}{dx}\right)^2 + \frac{3}{2}z\left(\frac{d^2 z}{dx^2}\right) - \frac{3}{4}z^2\left(\frac{dz}{dx}\right) - \frac{x}{4t}\left(\frac{dz}{dx}\right) - \frac{z}{4t}. \quad (4.3)$$

Utilizing (4.2) and (4.3), we can express the similarity transformation pair as follows:

$$\gamma = \frac{x}{t^{1/3}}, \quad (4.4a)$$

$$z = t^{-1/3} g(\gamma). \quad (4.4b)$$

Equations (4.2) and (4.3), when expressed in terms of the new variables γ and $g(\gamma)$ respectively, take the following form:

$$\frac{dg}{d\gamma} = \frac{1}{2}g^2 - \frac{\gamma}{6}, \quad (4.5)$$

$$\frac{d^3 g}{d\gamma^3} - \frac{3}{2}\left(\frac{dg}{d\gamma}\right)^2 - \frac{3}{2}g\frac{d^2 g}{d\gamma^2} = \gamma\frac{dg}{d\gamma} - \frac{3}{4}g^2\frac{dg}{d\gamma} - \frac{g}{4}. \quad (4.6)$$

Now, (4.5) bears a striking resemblance to the well-known “Riccati equation.” The standard form of the Riccati equation is expressed as

$$\frac{dg}{d\gamma} = a(\gamma)g^2(\gamma) + b(\gamma)g(\gamma) + c(\gamma). \quad (4.7)$$

Now, we shall proceed with the requisite substitution

$$g(\gamma) = -\frac{1}{a(\gamma)U} \frac{dU}{d\gamma}, \quad (4.8)$$

and readily obtain the following equation from (4.5)–(4.8):

$$\frac{d^2U}{d\gamma^2} - \frac{1}{12} \gamma U = 0. \quad (4.9)$$

Equation (4.9) conforms to the familiar “Airy’s equation/Stokes equation” which yields the following solution:

$$U(\gamma) = c_1 Ai\left(\frac{\gamma}{W}\right) + c_2 Bi\left(\frac{\gamma}{W}\right), \quad (4.10)$$

where c_1 and c_2 stand as arbitrary constants and $W = 2^{2/3} \cdot 3^{1/3}$. To streamline, we opt for $c_1 = c_2 = 1$. Consequently, we can easily ascertain the value of $g(\gamma)$ from (4.8). Utilizing (4.4a), (4.4b), the ultimate solution z to (4.2) unfolds as

$$z(x, t) = -t^{1/3} \frac{\left(\frac{1}{W}\right) [Ai'(W_0) + Bi'(W_0)]}{\frac{1}{2} [Ai(W_0) + Bi(W_0)]}. \quad (4.11)$$

where $W_0 = (x/t^{1/3})/W$.

The above solution expressed with respect to x_1 can be articulated as follows:

$$z_1(x, t) = -t^{1/3} \frac{\left(\frac{1}{W}\right) [Ai'(W_1) + Bi'(W_1)]}{\frac{1}{2} [Ai(W_1) + Bi(W_1)]}, \quad (4.12)$$

where $W_1 = ((x + x_1)/t^{1/3})/W$.

With $u = -z_{1x}$, the solution for the CKdV equation is given here as under

$$u(x, t) = \left[\left(\frac{2}{3}\right)^{1/3} \left(\frac{1}{t}\right) (x + x_1) \left(\frac{1}{W}\right) \right] - \left[\frac{\left(\frac{2}{3}\right)^{1/3} \left(\frac{1}{t}\right)^{2/3} \left(\frac{1}{W}\right) (Ai'[W_1] + Bi'[W_1])^2}{(Ai[W_1] + Bi[W_1])^2} \right]. \quad (4.13)$$

The solution (4.13) indicates the single soliton solution of the CKdV (3.8).

5 Theorem of Permutability (TP)

The permutability theorem, as delineated in reference [62], posits that the sequence of two consecutive Bäcklund trans-

formations, characterized by distinct parameters x_1 and x_2 , is commutative. Specifically, when these transformations are sequentially applied to an initial solution z , yielding a resultant solution z_{12} , the outcome remains invariant regardless of the order of their application. Thus, the consequent solution z_{12} is identical whether the transformations are applied first with x_1 followed by x_2 , or vice versa, illustrating the commutative nature of the BT [62]. Namely, if

$$z \xrightarrow{x_1} z_1 \xrightarrow{x_2} z_{12}, \quad (5.1)$$

and

$$z \xrightarrow{x_2} z_2 \xrightarrow{x_1} z_{21}, \quad (5.2)$$

then, of course, the TP requires

$$z_{12} = z_{21}. \quad (5.3)$$

Quantitatively, using the BT in (4.1a) and exploiting TP, the two-soliton solution of the CKdV equation reads as follows:

$$z_{12} = z + \frac{x_1 - x_2}{3t(z_1 - z_2)}. \quad (5.4)$$

Like z_1 (see (4.12)), z_2 can be expressed as follows:

$$z_2(x, t) = -t^{1/3} \left(\frac{1}{W}\right) \frac{[Ai'[W_2] + Bi'[W_2]]}{\frac{1}{2} [Ai[W_2] + Bi[W_2]]}, \quad (5.5)$$

where $W_2 = ((x + x_2)/t^{1/3})/W$.

It is crucial to note that (5.4) preserves symmetry upon interchanging indices 1 and 2, aligning with the stipulations of the TP, demonstrating that

$$u_{12}(x, t) = -\frac{\partial z_{12}(x, t)}{\partial x}. \quad (5.6)$$

The two-soliton solution of (3.8) can be found by making use of (5.6) and setting $w = 0$ in (5.4) which then reads as

follows:

where z_1 is the first soliton solution.

$$u_{12}(x, t) = \frac{(x_1 - x_2) \left[-\frac{C \left(\frac{D \cdot A_1 + D \cdot B_1}{2^{1/3} \cdot 3^{1/3}} \right)}{A_1 + B_1} + \frac{C \left(\frac{E \cdot A_2 + E \cdot B_2}{2^{1/3} \cdot 3^{1/3}} \right)}{A_2 + B_2} + \frac{C(A'_1 + B'_1) \left(\left(\frac{1}{t} \right)^{1/3} \frac{A'_1 + \left(\frac{1}{t} \right)^{1/3} B'_1}{2^{2/3} \cdot 3^{1/3}} \right)}{(A_1 + B_1)^2} - \frac{C(A'_2 + B'_2) \left(\left(\frac{1}{t} \right)^{1/3} \frac{A'_2 + \left(\frac{1}{t} \right)^{1/3} B'_2}{2^{2/3} \cdot 3^{1/3}} \right)}{(A_2 + B_2)^2} \right]}{3t \left(-\frac{C(A'_1 + B'_1)}{A_1 + B_1} + \frac{C(A'_2 + B'_2)}{A_2 + B_2} \right)^2}, \quad (5.7)$$

with

$$\begin{aligned} A_1 &= Ai \left(\left(\frac{1}{t} \right)^{1/3} (x + x_1) \right), B_1 = Bi \left(\left(\frac{1}{t} \right)^{1/3} (x + x_1) \right), \\ A_2 &= Ai \left(\left(\frac{1}{t} \right)^{1/3} (x + x_2) \right), B_2 = Bi \left(\left(\frac{1}{t} \right)^{1/3} (x + x_2) \right), \\ A'_1 &= Ai' \left(\left(\frac{1}{t} \right)^{1/3} (x + x_1) \right), B'_1 = Bi' \left(\left(\frac{1}{t} \right)^{1/3} (x + x_1) \right), \\ A'_2 &= Ai' \left(\left(\frac{1}{t} \right)^{1/3} (x + x_2) \right), B'_2 = Bi' \left(\left(\frac{1}{t} \right)^{1/3} (x + x_2) \right), \\ C &= \left(\frac{2}{3} \right)^{1/3} \left(\frac{1}{t} \right)^{1/3}, D = \left(\frac{1}{t} \right)^{2/3} (x + x_1), E = \left(\frac{1}{t} \right)^{2/3} (x + x_2). \end{aligned}$$

The readers are encouraged to see our earlier published article [46], where we have provided schematic diagrams for constructing two- and three-soliton solutions.

Adopting the procedure outlined above and after generalization, the solution at the n^{th} step can be expressed as

$$z(n) = z_{(n-2)} + \frac{x_{(n-1)} - x_{(n)}}{3t \left(z_{(n-1)} - z'_{(n-1)} \right)}, \quad (5.8)$$

$$z(n) = z_{\{k_1, k_2, \dots, k_n\}} = z_{12 \dots n}, \quad (5.9)$$

$$z'_{(n)} = z_{\{k_1, \dots, k_{n-1}, k_{n+1}\}} = z_{12 \dots n-1, n+1}, \quad (5.10)$$

and

$$z = z_{(0)} = 0. \quad (5.11)$$

The solution for three solitons can similarly be expressed as

$$z_{123} = z_1 + \frac{x_2 - x_3}{3t(z_{12} - z_{13})}, \quad (5.12)$$

6 Results and Discussion

Here, we discuss the study of nonlinear IASWs in a cylindrical geometry in electron trapping in a quantizing field and finite electron temperature. It must be mentioned that the chosen plasma parameters are representative of white dwarf stars, i.e., $n_e = 10^{26} \text{ cm}^{-3}$ and $B_0 = 10^{10} \text{ G}$ [63, 64]. There have been hundreds of observations of pulsating white dwarfs. This has led to the emergence of the field of white dwarf asteroseismology. The pulsations are attributed to the presence of gravity mode, also known as g-mode, according to the reasonably well-established theoretical model. Besides the g-modes, the theory also predicts the p-modes that are surmised to be electrostatic acoustic excitations. It is, therefore, imperative to investigate the acoustic type modes in the white dwarf stars [65–70]. The CKdV equation allows the formation of a compressive IA solitary structure for our system. The solutions of the CKdV equation are very different from the planar KdV equation. The one soliton solution of the CKdV equation is plotted for our physical system. The two- and three-soliton solutions are also given, and finally, the procedure to obtain N-soliton solutions for the CKdV equation is presented. Plotting the multi-soliton solutions is a daunting task due to the functional form of the solution for the CKdV equation. However, pursuing this direction is desirable and may involve expressing Airy's functions in terms of more tractable mathematical functions.

6.1 Variation of the Quantizing Magnetic Field Parameter η

Figure 1 investigates the effect of quantizing magnetic field on cylindrical ion-acoustic soliton (CIAS) structure by varying the parameter η , keeping the finite electron temperature fixed. The interested reader is referred to the recent work of Shah et al. [44], where the authors have used the pictorial representation to explain the effect of Landau quantization

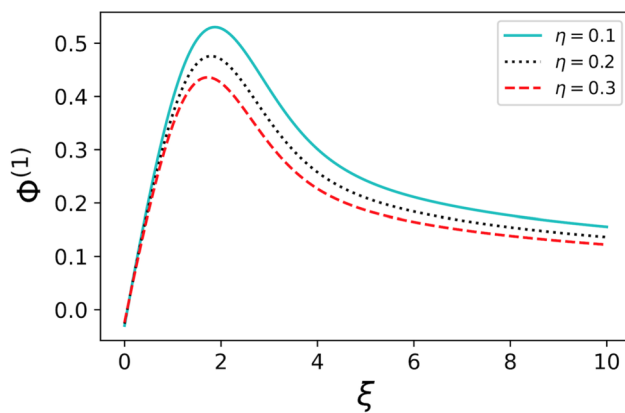


Fig. 1 Variation of the CKdV soliton for different values of quantizing magnetic field parameter η with $T = 0.2$

on electron trapping. It is found in our case that the amplitude of the CIAS mitigates with the increasing values of η .

6.2 Variation of the Finite Electron Temperature T

Figure 2 explores the effect of finite electron temperature T on the propagation attributes of the CIAS by keeping the quantizing parameter η fixed. It is observed that the increasing values of the finite electron temperature T augment the amplitude of the CIAS. This implies that the increasing T enhances the restoring force on the ion wave, which amplifies the amplification of the amplitude of the CIAS.

6.3 Limiting Cases

6.3.1 Variation of Finite Electron Temperature T in the Absence of Landau Quantization Parameter η

Figure 3 explores how the amplitude of the CIASs varies with the increasing values of T in the absence of η . Like Fig. 2, the finite electron temperature's rising values enhance

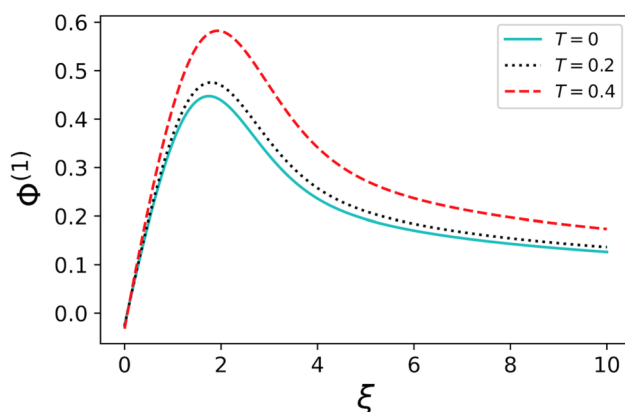


Fig. 2 Variation of the CKdV soliton for increasing values of the finite electron temperature T with $\eta = 0.2$

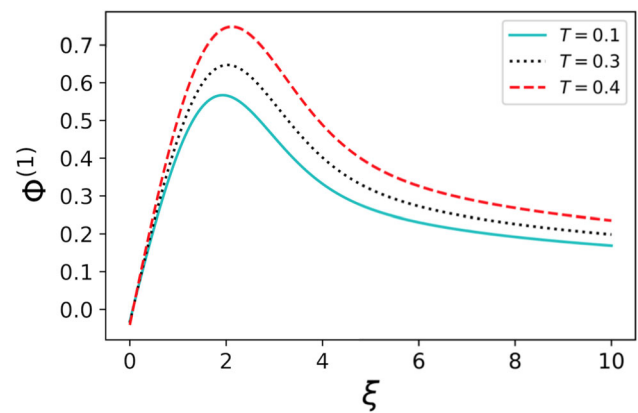


Fig. 3 Variation of finite electron temperature T in the absence of Landau quantization parameter $\eta = 0$

the CIASs' amplitude. However, the values of the perturbed potential are higher in this case. Once again, it alludes to the fact that the Landau quantization parameter, η , mitigates the amplitude of the CIASs.

6.3.2 Variation of Landau Quantization Parameter η in the Absence of Finite Electron Temperature T

Figure 4 examines the impact of increasing values of the quantizing magnetic field parameter η in the absence of finite electron temperature T . The analysis reveals that the pattern resembles the one depicted in Fig. 1. However, the decrease in the amplitude of the CIASs in this case is more significant. This again confirms that the amplitude of the soliton gets enhanced in the presence of a finite electron temperature T .

To facilitate the reader, we finally show all the cases in Fig. 5, which confirms all the earlier discussions regarding the effects of finite electron temperature T and the Landau quantization parameter η on the propagation attributes of

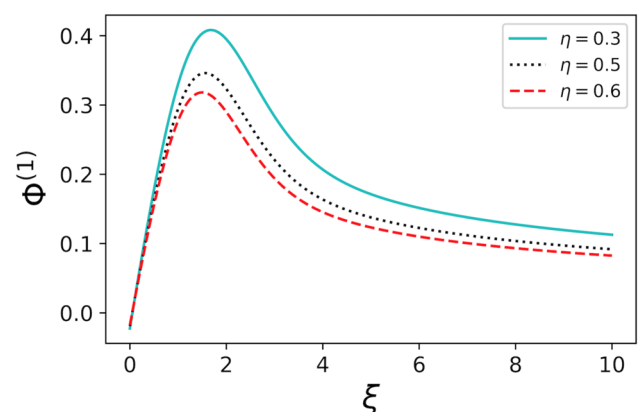


Fig. 4 Variation of Landau quantization parameter η in the absence of finite electron temperature $T = 0$

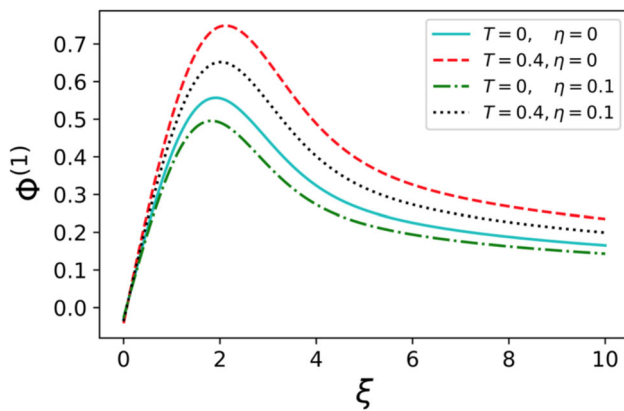


Fig. 5 Comparison of the CKdV soliton for different values of T and η

the CIASs. Note that the normalized variable ξ has been normalized by Fermi screening length λ_{TF} whose value for the range of white dwarf parameters turns out to be in nano-centimeters, and therefore, looking at the plots, it is easy to tell that the scale of formation of these structures varies from 1 – 10 nano-centimeters.

7 Conclusion

To summarize, we have examined the nonlinear ion-acoustic waves propagating in a cylindrical geometry with quantum mechanically trapped electrons in finite electron temperature and Landau quantization in a cylindrical geometry in an unmagnetized, collisionless plasma. Employing the multiple time scale analysis, we have derived cylindrical Korteweg-de Vries (CKdV) for our physical model. We have furnished single- and multi-soliton solutions of the CKdV equation using the Backlund transformation and have also given a general expression of solutions up to the n th order. It has been found that the CKdV equation admits solutions that are very different from the planar KdV equation and relatively much more complex in their mathematical form. We have found that the finite electron temperature and Landau quantization of the trapped electrons alter the propagation of nonlinear ion-acoustic waves in dense plasmas such as white dwarf stars. We have also underlined the obstacles that are encountered to graphically illustrate the multi-soliton interaction of the ion-acoustic solitons in a cylindrical geometry. We have highlighted the importance of investigating ion-acoustic mode with special reference to p-type mode in white dwarf stars where the ions provide the inertia whereas the restoring force comes from the degenerate electrons. This is a less explored area in plasma physics, and we hope the present investigation will spark an interest in the community to pursue this research direction.

8 Future Work

As an extension of this work, the Bäcklund transformation method is expected to give good results for some other evolution equations, including cylindrical Kadomtsev-Petviashvili (KP)-type equations, cylindrical Zakharov-Kuznetsov (ZK)-type equations, and cylindrical nonlinear Schrödinger (NLS)-type equations [71–73], which are used in studying and analyzing different nonlinear phenomena, whether in different plasma systems, fluid mechanics, or optical fibers.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of Interest The authors declare no competing interests.

Use of Artificial Intelligence (AI) Tools Declaration The authors declare they have not used AI tools in the creation of this article.

References

1. P. Shukla, B. Eliasson, Colloquium: nonlinear collective interactions in quantum plasmas with degenerate electron fluids. *Rev. Mod. Phys.* **83**(3), 885 (2011)
2. D.B. Melrose, *Quantum plasmadynamics: unmagnetized plasmas*, vol. 735 (Springer, 2008)
3. P.K. Shukla, B. Eliasson, Nonlinear theory for a quantum diode in a dense fermi magnetoplasma. *Phys. Rev. Lett.* **100**(3), 036801 (2008)
4. L. Ang, P. Zhang, Ultrashort-pulse child-Langmuir law in the quantum and relativistic regimes. *Phys. Rev. Lett.* **98**(16)(2007)
5. K. Becker, K. Schoenbach, J. Eden, Microplasmas and applications. *J. Phys. D Appl. Phys.* **39**(3), R55 (2006)
6. G. Shpatkovskaya, Semiclassical model of a one-dimensional quantum dot. *J. Exp. Theor. Phys.* **102**, 466–474 (2006)
7. R.M. Abolfath, A. Petukhov, I. Žutić, Piezomagnetic quantum dots. *Phys. Rev. Lett.* **101**(20), 207202 (2008)
8. M. Marklund, G. Brodin, L. Stenflo, C.S. Liu, New quantum limits in plasmonic devices. *Europhys. Lett.* **84**(1), 17006 (2008)

9. M.I. Stockman, Nanoplasmonics: the physics behind the applications. *Phys. Today* **64**(2), 39–44 (2011)
10. W.L. Barnes, A. Dereux, T.W. Ebbesen, Surface plasmon subwavelength optics. *nature* **424**(6950), 824–830 (2003)
11. P. Norreys, F. Beg, Y. Sentoku, L. Silva, R. Smith, R. Trines, Intense laser-plasma interactions: new frontiers in high energy density physics. *Phys. Plasmas* **16**(4), 041002 (2009)
12. M. Marklund, P.K. Shukla, Nonlinear collective effects in photon-photon and photon-plasma interactions. *Rev. Mod. Phys.* **78**(2), 591 (2006)
13. H. Azechi et al., Present status of the FIREX programme for the demonstration of ignition and burn. *Plasma Phys. Controlled Fusion* **48**(12B), B267 (2006)
14. R. Follett, J. Delettrez, D. Edgell, R. Hennen, J. Katz, J. Myatt, D. Froula, Plasma characterization using ultraviolet Thomson scattering from ion-acoustic and electron plasma waves. *Rev. Sci. Instrum.* **87**(11), 11E401 (2016)
15. H. Van Horn, Dense astrophysical plasmas. *Science* **252**(5004), 384–389 (1991)
16. G. Chabrier, F. Douchin, A. Potekhin, Dense astrophysical plasmas. *J. Phys.: Condens. Matter* **14**(40), 9133 (2002)
17. T. Guillot, Interiors of giant planets inside and outside the solar system. *science* **286**(5437), 72–77 (1999)
18. D. Lai, Matter in strong magnetic fields. *Rev. Mod. Phys.* **73**(3), 629 (2001)
19. K.T.A.L. Burm, Plasma: the fourth state of matter. *Plasma Chem. Plasma Process.* **32**, 401–407 (2012)
20. S.L. Shapiro, S.A. Teukolsky, *Black holes, white dwarfs, and neutron stars: the physics of compact objects* (John Wiley & Sons 2008)
21. P.K. Shukla, B. Eliasson, Nonlinear aspects of quantum plasma physics. *Phys. Usp.* **53**(1), 51 (2010)
22. E. Madelung, Quantum theory in hydrodynamical form. *z. Phys.* **40**, 322 (1927)
23. F. Haas, *Quantum plasmas: an hydrodynamic approach*, vol. 65 (Springer Science & Business Media, 2011)
24. S.V. Vladimirov, Y.O. Tyshetskiy, On description of a collisionless quantum plasma. *Phys. Usp.* **54**(12), 1243 (2011)
25. E.P. Wigner, On the quantum correction for thermodynamic equilibrium. Part I: Physical Chemistry. Part II: Solid State Physics, 110–120 (1997)
26. G. Manfredi, F. Haas, Self-consistent fluid model for a quantum electron gas. *Phys. Rev. B* **64**(7), 075316 (2001)
27. D. Bohm, D. Pines, A collective description of electron interactions: Iii. coulomb interactions in a degenerate electron gas. *Phys. Rev.* **92**(3), 609 (1953)
28. N. Tsintsadze, L. Tsintsadze, Novel quantum kinetic equations of the fermi particles. *Europhys. Lett.* **88**(3), 35001 (2009)
29. A. Gurevich, L. Pitaevskii, Recombination coefficient in a dense low-temperature plasma. *Sov. Phys. JETP* **19**(4), 870 (1964)
30. H. Shah, M. Qureshi, N. Tsintsadze, Effect of trapping in degenerate quantum plasmas. *Phys. Plasmas* **17**(3), 032312 (2010)
31. H. Shah, W. Masood, M. Qureshi, N. Tsintsadze, Effects of trapping and finite temperature in a relativistic degenerate plasma. *Phys. Plasmas* **18**(10) (2011)
32. H. Shah, M. Iqbal, N. Tsintsadze, W. Masood, M. Qureshi, Effect of trapping in a degenerate plasma in the presence of a quantizing magnetic field. *Phys. Plasmas* **19**(9), 092304 (2012)
33. M. Iqbal, H. Shah, W. Masood, N. Tsintsadze, Nonlinear ion acoustic waves in a relativistic degenerate plasma with landau diamagnetism and electron trapping. *Eur. Phys. J. D* **72**, 1–9 (2018)
34. W. Masood, M.I. Shaukat, A.M. Mirza, Obliquely propagating quasi one dimensional electrostatic solitary structures in dense magnetoplasmas with trapped electrons. *Astrophys. Space Sci.* **352**, 621–626 (2014)
35. W. Masood, M.I. Shaukat, H. Shah, A.M. Mirza, Solitary structures in a spatially nonuniform degenerate plasma in the presence of quantizing magnetic field. *Phys. Plasmas* **22**(3), 032305 (2015)
36. M.I. Shaukat, W. Masood, H. Shah, M. Iqbal, A.M. Mirza, Finite amplitude nonlinear drift waves in a spatially inhomogeneous degenerate plasma with landau quantization and electron temperature corrections. *Phys. Plasmas* **23**(10), 102310 (2016)
37. W. Masood, H. Shah, M. Qureshi, Trapping in quantum plasmas: a review. *Rev. Mod. Plasma Phys.* **6**(1), 11 (2022)
38. Z. Iqbal, H. Shah, M. Qureshi, W. Masood, A. Fayyaz, Nonlinear dynamical analysis of drift ion acoustic shock waves in electron-positron-ion plasma with adiabatic trapping. *Results Phys.* **41**, 105948 (2022)
39. H.A. Shah, T. Asam, Z. Iqbal, A. Bukhari, W. Masood, L.Z. Kahlon, Effect of trapping in coupled kinetic Alfvén-acoustic waves in a partially degenerate plasma with quantizing magnetic field. *Physica Scripta* (2023)
40. W. Masood, N. Hamid, S. Ullah, A. ur Rahman, H. Shah, H.A. Alyousef, S. El-Tantawy, Formation of acoustic nonlinear structures in non-Maxwellian trapping plasmas. *Phys. Fluids* **34**(5), 053113 (2022)
41. U. Erkaev, G. Gulyamov, J. Mirzaev, R. Rakhimov, N. Sayidov, Calculation of the Fermi-Dirac function distribution in two-dimensional semiconductor materials at high temperatures and weak magnetic fields. *Nano* **16**(09), 2150102 (2021)
42. U. Erkaev, G. Gulyamov, R. Rakhimov, A new method for determining the bandgap in semiconductors in presence of external action taking into account lattice vibrations. *Indian J. Phys.* **96**(8), 2359–2368 (2022)
43. U.I. Erkaev, U.M. Negmatov, R.G. Rakhimov, J.I. Mirzaev, N.A. Sayidov et al., Influence of a quantizing magnetic field on the Fermi energy oscillations in two-dimensional semiconductors. *Int. J. Appl. Sci. Eng.* **19**(2), 1–8 (2022)
44. S. Shah, W. Masood, M. Siddiq, H. Rizvi, Nonlinear ion acoustic waves in dense magnetoplasmas: analyzing interaction solutions of the KdV equation using Wronskian formalism for electron trapping with Landau diamagnetism and thermal excitations. *Chaos, Solitons & Fractals* **181**(2024)
45. M.T. Asam, H.A. Shah, W. Masood, S.A. Bukhari, Damped quantum drift Zakharov-Kuznetsov equation for a dissipative inhomogeneous partially degenerate plasma with quantizing magnetic field. *Phys. Scr.* **98**(2023)
46. M.S. Tariq, W. Masood, M. Siddiq, S. Asghar, B. Alotaibi, S.M. Ismael, S. El-Tantawy, Bäcklund transformation for analyzing a cylindrical Korteweg–De Vries equation and investigating multiple soliton solutions in a plasma. *Phys. Fluids* **35**(10) (2023)
47. S. El-Tantawy, P. Carbonaro, Nonplanar ion-acoustic solitons collision in Xe⁺-F-SF₆⁻ and Ar⁺-F-SF₆⁻ plasmas. *Phys. Lett. A* **380**(18–19), 1627–1634 (2016)
48. S. El-Tantawy, W. Moslem, R. Sabry, S. El-Labany, M. El-Metwally, R. Schlickeiser, Nonplanar solitons collision in ultracold neutral plasmas. *Phys. Plasmas* **20**(9) (2013)
49. S.A. Almutlak, M. Khalid, S.A. El-Tantawy, Nonplanar ion-acoustic solitary and cnoidal waves in a non-Maxwellian plasma: study on nonplanar (modified) Kawahara equation. *J. Low Freq. Noise Vib. Act. Control* **43**(2), 577–594 (2024)
50. S. El-Tantawy, L.S. El-Sherif, A. Bakry, W. Alhejaili, A.-M. Wazwaz, On the analytical approximations to the nonplanar damped Kawahara equation: cnoidal and solitary waves and their energy. *Phys. Fluids* **34**(11) (2022)
51. M. Alharthi, R. Alharbey, S. El-Tantawy, Novel analytical approximations to the nonplanar Kawahara equation and its plasma applications. *Eur. Phys. J. Plus* **137**(10), 1–10 (2022)
52. S. El-Tantawy, S.A. Shan, N. Mustafa, M.H. Alshehri, F.Z. Duraihem, N.B. Turki, Homotopy perturbation and Adomian decomposition

- sition methods for modeling the nonplanar structures in a bi-ion ionospheric superthermal plasma. *Eur. Phys. J. Plus* **136**(5), 561 (2021)
53. S. El-Tantawy, A.H. Salas, M.R. Alharthi, On the analytical and numerical solutions of the damped nonplanar Shamel Korteweg–De Vries burgers equation for modeling nonlinear structures in strongly coupled dusty plasmas: multistage homotopy perturbation method. *Phys. Fluids* **33**(4) (2021)
 54. S.A. Almutlak, S. El-Tantawy, On the approximate solutions of a damped nonplanar modified Korteweg–De Vries equation for studying dissipative cylindrical and spherical solitons in plasmas. *Results Phys.* **23**, 104034 (2021)
 55. B.S. Kashkari, S. El-Tantawy, A.H. Salas, L. El-Sherif, Homotopy perturbation method for studying dissipative nonplanar solitons in an electronegative complex plasma. *Chaos, Solitons & Fractals* **130**, 109457 (2020)
 56. S. El-Tantawy, T. Aboelenen, S.M. Ismaeel, Local discontinuous Galerkin method for modeling the nonplanar structures (solitons and shocks) in an electronegative plasma. *Phys. Plasmas* **26**(2) (2019)
 57. A. Nakamura, Bäcklund transformation of the cylindrical KdV equation. *J. Phys. Soc. Jpn.* **49**(6), 2380–2386 (1980)
 58. M. Shohaib, W. Masood, M. Siddiq, H.A. Alyousef, S.A. El-Tantawy, Formation of electrostatic solitary and periodic waves in dusty plasmas in the light of voyager 1 and 2 spacecraft and Freja satellite observations. *J. Low Freq. Noise Vib. Act. Control* **41**(3), 896–909 (2022)
 59. S. Maxon, J. Viecelli, Cylindrical solitons. *The. Phys. Fluids* **17**(8), 1614–1616 (1974)
 60. N. Hershkovitz, T. Romesser, Observations of ion-acoustic cylindrical solitons. *Phys. Rev. Lett.* **32**(11), 581 (1974)
 61. A. Nakamura, The Miura transform and the existence of an infinite number of conservation laws of the cylindrical KdV equation. *Phys. Lett. A* **82**(3), 111–112 (1981)
 62. A. Das, *Integrable models*, vol. 30 (World scientific, 1989)
 63. D. Koester, G. Chanmugam, Physics of white dwarf stars. *Rep. Prog. Phys.* **53**(7), 837 (1990)
 64. S. Ghosh, G. Lakhina, Anomalous width variation of rarefactive ion acoustic solitary waves in the context of auroral plasmas. *Nonlinear Process. Geophys.* **11**(2), 219–228 (2004)
 65. B. Eliasson, P. Shukla, The formation of electrostatic shocks in quantum plasmas with relativistically degenerate electrons. *Europhys. Lett.* **97**(1), 15001 (2011)
 66. W. Masood, B. Eliasson, Electrostatic solitary waves in a quantum plasma with relativistically degenerate electrons. *Phys. Plasmas* **18**(3) (2011)
 67. A.-u.-R. Ata-ur Rahman, W. Masood, B. Eliasson, A. Qamar, Three dimensional electrostatic solitary waves in a dense magnetoplasma with relativistically degenerate electrons. *Phys. Plasmas* **20**(9) (2013)
 68. H. Shah, W. Masood, M. Asim, M. Qureshi, Drift solitary structures in inhomogeneous degenerate quantum plasmas with trapped electrons. *Astrophys. Space Sci.* **350**, 615–622 (2014)
 69. R. Jahangir, W. Masood, M. Siddiq, N. Batool, K. Saleem, Interaction of fast magnetoacoustic solitons in dense plasmas. *Phys. Plasmas* **22**(9) (2015)
 70. A. Sabeen, H. Shah, W. Masood, M. Qureshi, Finite amplitude solitary structures of coupled kinetic Alfvén-acoustic waves in dense plasmas. *Astrophys. Space Sci.* **355**, 225–232 (2015)
 71. S. El-Tantawy, R. Alharbey, A.H. Salas, Novel approximate analytical and numerical cylindrical rogue wave and breathers solutions: an application to electronegative plasma. *Chaos, Solitons & Fractals* **155**, 111776 (2022)
 72. S. El-Tantawy, A.H. Salas, H.A. Alyousef, M. Alharthi, Novel approximations to a nonplanar nonlinear Schrödinger equation and modeling nonplanar rogue waves/breathers in a complex plasma. *Chaos, Solitons & Fractals* **163**(2022)
 73. S. El-Tantawy, T. Aboelenen, Simulation study of planar and nonplanar super rogue waves in an electronegative plasma: local discontinuous Galerkin method. *Phys. Plasmas* **24**(5) (2017)

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